FUNCTIONS & GRAPHS 1 — UNDERSTANDING FUNCTIONS

Canonical Practice Bank with Full Solutions

CORE → CHALLENGE → ADVANCED

Practice principle

Independent questions are structurally separated from the Teaching Material. Use Cambridge composition notation $fg(x) = f(g(x))$. For Challenge and Advanced, always check function structure, domain/range validity and the meaning of the final result.

PART I — CORE

C1 — Function or not?

Relation A: $-2 \mapsto 1$, $0 \mapsto 1$, $3 \mapsto 5$.
Relation B: $1 \mapsto 2$, $1 \mapsto 4$, $3 \mapsto 6$.
(a) Which relation is a function? (b) Is A one-one or many-one? (c) Why does B fail?

C2 — Function notation

Let $f(x) = 2x^2 - 3x + 1$. Find (a) $f(-2)$, (b) $f(a)$, (c) $f(x+1)$.

C3 — Output given, find the input

Let $g(x) = 4x - 7$ and $g(a) = 21$. Find a.

C4 — Domain restrictions

State the real domain of: (a) $f(x) = \dfrac{5}{x-2}$; (b) $g(x) = \sqrt{2x+6}$; (c) $h(x) = x^3 - 4x + 1$. Explain each restriction.

C5 — Range from a restricted domain

Let $f(x) = x^2 + 1$ with $-2 \le x \le 3$. Find the range.

C6 — Domain, range, one-one

A relation consists of $(-2, 3), (0, 1), (4, 3), (5, 6)$. Find the domain, range, whether it is a function, and whether it is one-one.

C7 — Which function is one-one?

Compare $f(x) = 3x - 2$ and $g(x) = x^2$ on $\mathbb{R}$. Which is one-one? Justify and give two different inputs showing why the other is not.

C8 — Composition order

Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find $fg(x)$ and $gf(x)$. Hence explain why $fg \neq gf$.

C9 — Simple inverse

Let $f(x) = 5x - 4$. (a) Find $f^{-1}(x)$. (b) Find $f^{-1}(11)$. (c) Interpret the result in terms of f.

C10 — Track a point through transformations

The point $(-1, 4)$ lies on $y = f(x)$. Find the corresponding point on (a) $y = f(x-3) + 2$; (b) $y = -f(x)$.

PART II — CHALLENGE

H1 — Domain and range from a graph description

A continuous graph of $y = f(x)$ begins at an *open* point $(-3, 5)$, falls to a *closed* minimum $(1, -2)$, then rises to a *closed* endpoint $(4, 3)$; everywhere else it stays below $y = 5$. Determine domain and range.

H2 — Same functions, different order

Let $f(x) = x - 1$ and $g(x) = \sqrt{5 - x}$. Find $fg(x)$ and its domain, then $gf(x)$ and its domain. Explain why the domains differ.

H3 — Rational inverse + domain/range

Let $f(x) = \dfrac{2x + 3}{x - 1}$. State the domain, find $f^{-1}(x)$, determine the range of f, then state the domain and range of f^{-1}.

H4 — Restricted quadratic inverse (revised)

Let $f(x) = (x + 1)^2 - 4$ with domain $x \leq -1$. (a) Explain why f is one-one. (b) Find its range. (c) Find $f^{-1}(x)$. (d) State the domain and range of the inverse.

H5 — Verify an inverse

Let $f(x) = 3x - 4$. A student claims $g(x) = \dfrac{x + 4}{3}$ is f^{-1}. Verify using both $fg(x)$ and $gf(x)$.

H6 — Reconstruct a transformation

Points $P = (1, 3)$ and $Q = (4, -1)$ on $y = f(x)$ become $P' = (5, 8)$ and $Q' = (8, 0)$ under $y = 2f(x - h) + k$. Find h, k, and the transformed equation.

H7 — Absolute value transformation

The graph of $y = f(x)$ crosses the x-axis at $x = -1$ and $x = 3$, lies below the axis for $-1 < x < 3$, and passes through $(0, -4)$ and $(4, 2)$. For $y = |f(x)|$, determine what happens to the intercepts, both points, and which part reflects.

H8 — Missing function

Let $f(x) = x + 4$ and suppose $gf(x) = 2x + 1$. Find $g(x)$.

H9 — Invert a mapping

A one-one function maps $-2 \mapsto 5$, $0 \mapsto 1$, $3 \mapsto 7$. Write the inverse mapping and state $\text{Dom}(f^{-1})$ and $\text{Range}(f^{-1})$.

H10 — Range given, find restriction

Let $f(x) = (x - 1)^2$ with domain $x \geq k$, where $k \geq 1$. Its range is $y \geq 9$. Find k and explain why $k \geq 1$ matters.

PART III — ADVANCED

A1 — Recover a missing function from a composite

Let $g(x) = \sqrt{x + 1}$. For $x \geq -4$, it is known that $gf(x) = x + 4$. Find $f(x)$ on this domain and verify the composition is valid.

A2 — Composition + inverse

Let $f(x) = 3x - 1$ and $g(x) = \dfrac{1}{x+2}$. (a) Find $fg(x)$. (b) State its domain. (c) Find $(fg)^{-1}(x)$. (d) State domain and range of the inverse. (e) Verify one inverse composition.

A3 — Parameter from inverse domain

Let $f(x) = (x-2)^2 + k$, $x \geq 2$. The domain of f^{-1} is $x \geq 5$. Find k, $f^{-1}(x)$, and the range of the inverse.

A4 — Reconstruct a combined transformation

On $y = f(x)$, $P = (1, 2)$, $Q = (3, -1)$, $R = (5, 4)$. Under $y = af(x-h)+k$, they become $P' = (4, 5)$, $Q' = (6, -4)$, $R' = (8, 11)$. Find h, a, k and check all three points.

A5 — Restricted quadratic inverse (original Phase 4 version)

Let $h(x) = (2x-5)^2 + 3$ with domain $x < \dfrac{5}{2}$. (a) Find the range. (b) Find $h^{-1}(x)$. (c) State domain and range of h^{-1}. (d) Explain why the negative square-root branch is required.

A6 — Parameter for composition compatibility

Let $f(x) = x^2 + a$ for $0 \leq x \leq 3$, and $g(x) = \sqrt{13 - x}$. Find all a for which $gf(x)$ is defined for every x in the domain of f. If a is as large as possible, find the range of gf.

A7 — Graph, inverse and transformation

A strictly increasing function f has domain $1 \leq x \leq 5$ and passes through $(1, -2), (3, 4), (5, 7)$; endpoint values are min/max. State the range of f, the domain/range of f^{-1}, three inverse points, then their images on $y = f^{-1}(x) + 2$.

A8 — Vertical vs horizontal transformations (revised)

A point (a, b) lies on $y = f(x)$. Compare $y = 2f(x)$, $y = f(2x)$ and $y = f(-2x)$. (a) Find the image of (a, b) under each. (b) Use $(4, 3)$. (c) Explain why they differ. (d) Describe $y = f(-2x)$ geometrically.

A9 — Composition domains with two restrictions

Let $f(x) = \dfrac{1}{x-2}$ and $g(x) = \sqrt{x+1}$. Find $fg(x)$ and its domain; then $gf(x)$ and its domain, written as intervals or inequalities.

A10 — Function reconstruction + inverse chain

Let $f(x) = ax+b$, $a \neq 0$, and $g(x) = x-2$. Given $f^{-1}(7) = 3$ and $fg(6) = 10$, find $f(x)$, $f^{-1}(x)$, $h(x) = f^{-1}(g(x))$, and the x for which $h(x) = 5$.

FULL SOLUTIONS — CORE

C1

$\boxed{\text{A is a function and is many-one.}}$ Relation B is not a function because input 1 has two outputs, 2 and 4.

C2

$f(-2) = 15$. Also $f(a) = 2a^2 - 3a + 1$. For $f(x+1)$: $2(x+1)^2 - 3(x+1) + 1 = \boxed{2x^2 + x}$.

C3

$4a - 7 = 21 \Rightarrow 4a = 28 \Rightarrow \boxed{a = 7}$.

C4

(a) $x \neq 2$. (b) $2x + 6 \geq 0 \Rightarrow \boxed{x \geq -3}$. (c) Polynomial: $\boxed{x \in \mathbb{R}}$.

C5

On $[-2, 3]$, minimum occurs at $x = 0$: $f = 1$. Maximum occurs at $x = 3$: $f = 10$. Hence $\boxed{1 \leq y \leq 10}$.

C6

Dom $= \{-2, 0, 4, 5\}$; Range $= \{1, 3, 6\}$. It is a function, but not one-one because -2 and 4 both map to 3.

C7

$f(x) = 3x - 2$ is one-one. $g(x) = x^2$ is many-one because $g(2) = g(-2) = 4$.

C8

$fg(x) = f(x^2) = \boxed{2x^2 + 1}$. $gf(x) = g(2x + 1) = \boxed{(2x + 1)^2}$. Order changes the process.

C9

Swap x, y: $x = 5y - 4 \Rightarrow y = (x + 4)/5$. Thus $\boxed{f^{-1}(x) = \dfrac{x + 4}{5}}$ and $f^{-1}(11) = 3$, meaning $\boxed{f(3) = 11}$.

C10

For $f(x - 3) + 2$: $(a, b) \mapsto (a + 3, b + 2)$, so $\boxed{(2, 6)}$. For $-f(x)$: $\boxed{(-1, -4)}$.

FULL SOLUTIONS — CHALLENGE

H1

Horizontal extent gives $\boxed{-3 < x \leq 4}$. The minimum -2 is included, while 5 occurs only at the excluded left endpoint, so $\boxed{-2 \leq y < 5}$.

H2

Since $fg = f \circ g$, $fg(x) = \sqrt{5 - x} - 1$, requiring $x \leq 5$. Meanwhile $gf(x) = \sqrt{6 - x}$, requiring $x \leq 6$. The different order changes the expression entering the square root.

H3

Domain: $x \neq 1$. Let $y = (2x+3)/(x-1)$: $x(y-2) = y+3$, hence $\boxed{f^{-1}(x) = \dfrac{x+3}{x-2}}$. Output 2 is impossible, so Range$(f) = \mathbb{R} \setminus \{2\}$. Thus Dom$(f^{-1}) = \mathbb{R} \setminus \{2\}$ and Range$(f^{-1}) = \mathbb{R} \setminus \{1\}$.

H4

The restriction $x \leq -1$ keeps only the left branch, so f is one-one. Vertex $(-1,-4)$ gives range $y \geq -4$. From $y+4 = (x+1)^2$ and $x+1 \leq 0$, choose the negative branch: $x+1 = -\sqrt{y+4}$. Hence $\boxed{f^{-1}(x) = -1-\sqrt{x+4}}$, with domain $x \geq -4$ and range $y \leq -1$.

H5

$f(g(x)) = 3((x+4)/3) - 4 = x$ and $g(f(x)) = ((3x-4)+4)/3 = x$. Therefore $\boxed{g = f^{-1}}$.

H6

Horizontal movement $1 \to 5$ and $4 \to 8$ gives $\boxed{h = 4}$. Vertical equation $2(3) + k = 8$ gives $\boxed{k = 2}$; check $2(-1)+2 = 0$. Thus $\boxed{y = 2f(x-4)+2}$.

H7

Zeros remain zeros: $x = -1, 3$. $(0,-4) \mapsto \boxed{(0,4)}$; $(4,2)$ stays $\boxed{(4,2)}$. Exactly the part with $-1 < x < 3$ reflects across the x-axis.

H8

$g(f(x)) = g(x+4) = 2x+1$. Let $u = x+4$, so $x = u-4$. Then $g(u) = 2(u-4)+1 = 2u-7$. Hence $\boxed{g(x) = 2x-7}$.

H9

Outputs $5, 1, 7$ are distinct, so f is one-one. Reverse arrows: $5 \mapsto -2$, $1 \mapsto 0$, $7 \mapsto 3$. Thus Dom$(f^{-1}) = \{1,5,7\}$ and Range$(f^{-1}) = \{-2,0,3\}$.

H10

On $x \geq k \geq 1$, the branch is increasing, so minimum output is $(k-1)^2$. Set $(k-1)^2 = 9$: $k = 4$ or -2. Since $k \geq 1$, $\boxed{k = 4}$.

FULL SOLUTIONS — ADVANCED

A1

$\sqrt{f(x)+1} = x+4$. Because $x \geq -4$, RHS is non-negative. Squaring gives $f(x)+1 = (x+4)^2$, so $\boxed{f(x) = x^2+8x+15}$. Also $f(x)+1 = (x+4)^2 \geq 0$, and $\sqrt{(x+4)^2} = x+4$ on the stated domain, so the composition is valid.

A2

$fg(x) = 3/(x+2) - 1 = \boxed{(1-x)/(x+2)}$, $x \neq -2$. Let $y = (1-x)/(x+2)$: $x(y+1) = 1-2y$, so $\boxed{(fg)^{-1}(x) = \dfrac{1-2x}{x+1}}$. Domain of inverse excludes -1; range excludes -2. Substitution verifies one inverse composition simplifies to x.

A3

Since $x \geq 2$, the minimum of f is k, so Range$(f) = [k, \infty)$. This is the inverse domain $[5, \infty)$, hence $\boxed{k = 5}$. Then $\boxed{f^{-1}(x) = 2 + \sqrt{x-5}}$, whose range is $y \geq 2$.

A4

Horizontal movement gives $\boxed{h = 3}$. Using P, Q: $2a + k = 5$ and $-a + k = -4$. Subtract to get $a = 3$, then $k = -1$. Thus $\boxed{y = 3f(x-3) - 1}$. Point R: $x = 5 \to 8$, $y = 4 \to 11$, consistent.

A5

Domain $x < 5/2$ implies $2x - 5 < 0$ and the vertex is excluded, so $h(x) > 3$. Thus range is $\boxed{y > 3}$. From $y - 3 = (2x-5)^2$ choose $2x - 5 = -\sqrt{y-3}$, giving $\boxed{h^{-1}(x) = \dfrac{5 - \sqrt{x-3}}{2}}$. Domain $x > 3$; range $y < 5/2$. The negative branch is forced by $2x - 5 < 0$.

A6

Need $13 - (x^2 + a) \geq 0$ for every $0 \leq x \leq 3$. Worst case $x^2 = 9$, so $9 + a \leq 13$, hence $\boxed{a \leq 4}$. At $a = 4$, $gf(x) = \sqrt{9 - x^2}$ on $[0, 3]$, giving $\boxed{0 \leq y \leq 3}$.

A7

Strictly increasing from endpoint values -2 to 7, so Range$(f) = [-2, 7]$. Hence Dom$(f^{-1}) = [-2, 7]$, Range$(f^{-1}) = [1, 5]$. Inverse points: $(-2, 1), (4, 3), (7, 5)$. Add 2 vertically: $\boxed{(-2, 3), (4, 5), (7, 7)}$.

A8

For $2f(x)$: $(a, b) \mapsto \boxed{(a, 2b)}$. For $f(2x)$: $(a, b) \mapsto \boxed{(a/2, b)}$. For $f(-2x)$: $(a, b) \mapsto \boxed{(-a/2, b)}$. Thus $(4, 3)$ maps to $(4, 6)$, $(2, 3)$ and $(-2, 3)$ respectively. The last is reflection in the y-axis plus horizontal compression by factor $1/2$.

A9

$fg(x) = 1/(\sqrt{x+1} - 2)$. Require $x \geq -1$ and denominator nonzero: $x \neq 3$. Thus $\boxed{[-1, \infty) \setminus \{3\}}$. For $gf(x) = \sqrt{1/(x-2) + 1} = \sqrt{(x-1)/(x-2)}$, require $(x-1)/(x-2) \geq 0$, $x \neq 2$. Sign analysis gives $\boxed{(-\infty, 1] \cup (2, \infty)}$.

A10

$f^{-1}(7) = 3$ means $f(3) = 7$. Also $g(6) = 4$ and $fg(6) = 10$ means $f(4) = 10$. So $3a + b = 7$, $4a + b = 10$, giving $a = 3, b = -2$. Hence $\boxed{f(x) = 3x - 2}$, $\boxed{f^{-1}(x) = \dfrac{x+2}{3}}$. Then $h(x) = f^{-1}(x - 2) = x/3$, so $h(x) = 5$ gives $\boxed{x = 15}$.

QUESTION-FAMILY MAP

Family	Core	Challenge	Advanced
Function meaning	C1	embedded	mixed
Notation/reverse value	C2,C3	—	A10
Domain/range	C4-C6	H1-H4,H10	A2,A3,A5,A6,A7,A9
One-one/inverse	C7,C9	H3-H5,H9,H10	A2,A3,A5
Composition	C8	H2,H5,H8	A1,A2,A6,A9,A10
Transformations	C10	H6,H7	A4,A7,A8
Reconstruction/parameter	C3	H6,H8,H10	A1,A3,A4,A6,A10

SAFE RANDOMIZATION RULES

- **Mappings:** for a valid function, every input has exactly one outgoing arrow; many inputs may share an output.
- **Rational functions:** generate denominator/excluded values first; avoid accidental cancellation unless intentionally testing a hole.
- **Square roots:** generate the radicand first, then derive the domain.
- **Restricted quadratics:** choose vertex, branch and restriction first; derive range and inverse from that branch.
- **Composition:** construct inner range and outer domain deliberately; enforce or deliberately violate compatibility.
- **Inverses:** generate the original one-one function first, then derive its inverse; never randomize the pair independently.
- **Transformations:** generate original features first, then map every feature with the same coordinate rule.

HINT LADDERS

Domain

1. Which inputs could make the rule invalid?
2. Check denominators, roots and context.
3. Denominator $\neq 0$; real radicand ≥ 0.

Inverse

1. What process does the function perform?
2. Reverse order and operation.
3. Write $y = f(x)$ and swap roles of x, y.

Composition

1. Which function acts first?
2. Draw input $\rightarrow$ inner $\rightarrow$ outer.
3. For fg, start with $g(x)$.

Transformation

1. Track one point.
2. Separate input-coordinate and output-coordinate effects.
3. For $Af(B(x-h))+K$, use the same coordinate map for every feature.

VISUAL ANTI-LEAK & GRAPH RELIABILITY

State 0 shows only given graph/data; State 1 highlights a relevant point/endpoint; State 2 may show a scan, one reflected point or one point-movement arrow; State 3 may reveal one intermediate; State 4 may show full solved domain/range, inverse graph, transformed graph or parameters.

Every graph must preserve axis scale, intercepts, turning points, endpoints, open/closed status, restrictions, symmetry and exact transformation/reflection of key points. Do not display a final transformed or inverse graph before the learner has solved the corresponding task.

PRACTICE-SITE METADATA HANDOFF

Recommended per-question fields include:

- `id`, `band`, `family`, `representation`, `target`;
- `functions`, `domain`, `range`, `isOneOne`, `compositionOrder`;
- `visibleProblemData`, `derivedData`, `solutionData`;
- `graphSpec`, `hintStages`, `prerequisites`, `randomizationConstraints`.

Keep visible problem data structurally separate from derived and solution data.